

The Ax-Grothendieck Theorem

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September 30, 2026

Abstract

We give a brief exposition of the Ax-Grothendieck theorem. The Ax-Grothendieck theorem states that an injective endomorphism of an algebraic variety over an algebraically closed field is necessarily surjective. We won't be concerned with the general result but see the theorem for $\mathbb{K}^n$ for any algebraically closed field $\mathbb{K}$ with special importance to the field $\mathbb{C}$. We will first look at a complex-analytic proof of the theorem over $\mathbb{C}$, following Rudin. We will then turn to an algebraic proof, where the Nullstellensatz and the idea of "spreading out" allow us to pass from arbitrary algebraically closed fields to finite fields, reducing the theorem ultimately to the elementary fact that an injective map of a finite set is surjective. Following this idea we will also see a beautiful result of automorphism of the affine n -space following [Serre]

Since we will be dealing with affine varieties, we will use the term morphism and polynomial map interchangeably. By a polynomial map we mean

$$P : k^n \longrightarrow k^n,$$

where

$$P = (P_1, P_2, \dots, P_n), \quad P_i \in k[x_1, x_2, \dots, x_n].$$

Since both notions coincide for affine algebraic varieties, for morphisms in our case we take the definition to be a polynomial map.

Some Motivation for the Theorem

Lets be honest: algebraic geometry is not for commoners. What we, as humans, truly understand is some analysis, some algebra, and mostly linear algebra. So I have tried to present some motivation along these lines.

1. Fundamental Theorem of Algebra

Let us first consider the case $n = 1$. It is easy to see that injectivity implies that our polynomial is non-constant, and then, by the Fundamental Theorem of Algebra, we are through. Thus, in a sense, one may think of the Ax-Grothendieck Theorem as a generalization of the Fundamental Theorem of Algebra.

2. Linear Algebra

Now, before all the algebraic geometers start yelling at me that an algebraic variety is not a vector space, let me be clear: in general, it is not. Indeed, suppose we take a singular curve C . If it were a vector space (say with addition $\oplus$), then the addition map

$$\begin{aligned} C \times C &\longrightarrow C, \\ (P, Q) &\longmapsto P \oplus Q \end{aligned}$$

would be a morphism of varieties. Let P_0 be a singular point of C . Then, for any $Q \in C$, the translation map

$$\begin{aligned} T_Q : C &\longrightarrow C, \\ P &\longmapsto P \oplus Q \end{aligned}$$

is a morphism (in fact, an isomorphism). Since isomorphisms preserve singular points, every point $P_0 \oplus Q$ is singular. As Q varies over C , these points exhaust C . Hence every point of C would be singular. We know that since $\text{Sing}(C)$ is a proper closed subset of C , we have

$$\dim(\text{Sing}(C)) < \dim(C).$$

Thus this is impossible. [\[Har\]](#)

On the other hand, k^n inherits both a vector-space structure and an algebraic-variety structure. Since we already have a vector-space structure, why not inspect it?

We are interested in polynomial maps

$$P : k^n \longrightarrow k^n.$$

If all the coordinate polynomials P_i are linear, then P is nothing but a linear map. From finite-dimensional linear algebra we know that

$$P \text{ injective} \iff P \text{ surjective}.$$

But if we enlarge our class of maps from linear maps to polynomial maps, the converse implication no longer holds. For example,

$$\begin{aligned} f : \mathbb{C} &\longrightarrow \mathbb{C}, \\ z &\longmapsto z^2 \end{aligned}$$

is surjective but not injective. Remarkably, however, the other implication survives:

$$\boxed{P \text{ injective} \implies P \text{ surjective}.}$$

And this is precisely the point of the Ax–Grothendieck Theorem. For completely arbitrary functions, even this implication need not hold (see [\[Tao\]](#)). Thus polynomial maps retain just enough of the rigidity of linear algebra for the finite-dimensional phenomenon

$$\text{“injective} \implies \text{surjective”}$$

to survive. Now let's get to the actual proofs of the theorem.

Proof via Complex Analysis

We will use complex analysis to give a proof of the Ax–Grothendieck theorem for $\mathbb{K}=\mathbb{C}$. We will use a lemma that would give us that $\det(Df(z)) \neq 0 \forall z \in \mathbb{C}^n$. Now many would be tempted to use the notorious Jacobian Conjecture. But as it stands today, it has been disproved for $n \geq 3$. So we have no hope of using it but fortunately we have a proof due to Rudin that saves us and which we will follow. We first need a basic lemma.

Lemma 1. *Let $\Omega \subseteq \mathbb{C}^n$ be an open set, and let*

$$f : \Omega \longrightarrow \mathbb{C}^n$$

be an injective holomorphic map. Then the Jacobian of f is non-degenerate; that is,

$$\det Df(z) \neq 0$$

for every $z \in \Omega$.

Proof. We use an argument of Rosay and proceed by induction on n .

For $n = 1$, the claim follows immediately from the Taylor expansion. Indeed, suppose that $f'(z_0) = 0$. Since f is non-constant, we may write

$$f(z) = f(z_0) + a_m(z - z_0)^m + \cdots, \quad m \geq 2, \quad a_m \neq 0.$$

Locally, such a map behaves like $z \mapsto z^m$, and hence cannot be injective. Therefore $f'(z) \neq 0$ for every $z \in \Omega$. Now suppose that $n > 1$ and that the result has been proved in dimension $n - 1$. Suppose, for contradiction, that $\det Df(z_0) = 0$ for some $z_0 \in \Omega$.

We first claim that

$$Df(z_0) = 0.$$

Suppose otherwise. Then there exist $1 \leq i, j \leq n$ such that

$$\frac{\partial f_i}{\partial z_j}(z_0) \neq 0.$$

After permuting the coordinates, we may assume that $i = j = 1$. Furthermore, by translating the source and target, we may assume

$$z_0 = 0 \quad \text{and} \quad f(0) = 0.$$

Define

$$h(z_1, \dots, z_n) = (f_1(z_1, \dots, z_n), z_2, \dots, z_n).$$

Then

$$Dh(0) = \begin{pmatrix} \frac{\partial f_1}{\partial z_1}(0) & \frac{\partial f_1}{\partial z_2}(0) & \cdots & \frac{\partial f_1}{\partial z_n}(0) \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix},$$

and hence

$$\det Dh(0) = \frac{\partial f_1}{\partial z_1}(0) \neq 0.$$

By the holomorphic inverse function theorem, h is locally biholomorphic at 0.

Consider $g = f \circ h^{-1}$. Then g is holomorphic and injective in a neighbourhood of the origin. Moreover, by the definition of h , the first coordinate of g is simply the first coordinate:

$$g(w_1, \dots, w_n) = (w_1, g_2(w_1, \dots, w_n), \dots, g_n(w_1, \dots, w_n)).$$

Thus g preserves the hyperplane $H = \{w_1 = 0\} \cong \mathbb{C}^{n-1}$. Restricting g to H gives an injective holomorphic map

$$\tilde{g} : (\mathbb{C}^{n-1}, 0) \longrightarrow (\mathbb{C}^{n-1}, 0),$$

given locally by

$$(w_2, \dots, w_n) \longmapsto (g_2(0, w_2, \dots, w_n), \dots, g_n(0, w_2, \dots, w_n)).$$

By the induction hypothesis, $\det D\tilde{g}(0) \neq 0$.

On the other hand, the Jacobian matrix of g at the origin has the block form

$$Dg(0) = \begin{pmatrix} 1 & 0 \\ * & D\tilde{g}(0) \end{pmatrix}.$$

Therefore

$$\det Dg(0) = \det D\tilde{g}(0) \neq 0.$$

But $g = f \circ h^{-1}$, so by the chain rule, $Dg(0) = Df(0) Dh^{-1}(0)$. Since $Dh^{-1}(0)$ is invertible, this implies $\det Df(0) \neq 0$, contrary to our assumption.

We have therefore proved that

$$\det Df(z_0) = 0 \implies Df(z_0) = 0.$$

Now consider the analytic set

$$Z = \{z \in \Omega : \det Df(z) = 0\}.$$

If Z is nonempty and $\det Df$ is not identically zero, then Z is an analytic hypersurface (more precisely, its regular part is locally of complex dimension $n - 1$). By what we have just proved, $Df(z) = 0, \forall z \in Z$. Hence

the restriction of f to every connected piece of the regular locus of Z has zero differential and is therefore locally constant. Since such a piece has positive dimension when $n > 1$, this contradicts the injectivity of f . If instead $\det Df \equiv 0$, then the implication proved above gives $Df \equiv 0$ on Ω , so f is locally constant, again contradicting injectivity. Thus Z must be empty. Consequently,

$$\boxed{\det Df(z) \neq 0}$$

for every $z \in \Omega$. □

As a corollary we get the following result which we state as remark.

Remark 1. Every injective holomorphic map

$$f : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

is an open map.

Proof. By the preceding lemma, $\det Df(z) \neq 0$ for every $z \in \mathbb{C}^n$. Hence, by the holomorphic inverse function theorem, f is a local biholomorphism at every point. In particular, f is an open map. □

We can now give Rudin's proof of the Ax–Grothendieck theorem over $\mathbb{C}$.

Theorem 1 (Ax–Grothendieck over $\mathbb{C}$). *Let*

$$P : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

be a polynomial map. If P is injective, then P is surjective.

Rudin's proof. Let

$$P : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

be an injective polynomial map. Let k be the field generated by $\mathbb{Q}$ and the coefficients of P . Thus P is defined over k . Let $k(z) = k(z_1, \dots, z_n)$ be the rational function field in n indeterminates over k . Inside $k(z)$, consider the subfield $k(P(z)) = k(P_1(z), \dots, P_n(z))$ generated by k and the components of $P(z)$. We claim that

$$\boxed{k(P(z)) = k(z)}.$$

Suppose otherwise. Since P is injective, the extension

$$k(z)/k(P(z))$$

is algebraic, and since we are in characteristic zero, it is separable. Passing, if necessary, to a normal closure, Galois theory produces a non-trivial conjugate of the generic point z over $k(P(z))$. Thus there is a non-trivial rational transformation

$$\phi(z) = \left(\frac{Q_1(z)}{R_1(z)}, \dots, \frac{Q_n(z)}{R_n(z)} \right)$$

such that $P(\phi(z)) = P(z)$. Now specialize z to a sufficiently generic point of $\mathbb{C}^n$, avoiding the poles of ϕ and the locus on which $\phi(z) = z$. Since k is countable, we may choose the coordinates sufficiently transcendental over k . We then obtain two distinct points of $\mathbb{C}^n$ having the same image under P , contradicting the injectivity of P .

Therefore $k(P(z)) = k(z)$.

Consequently, for every $j = 1, \dots, n$, there exist relatively prime polynomials $Q_j, R_j \in k[w_1, \dots, w_n]$ such that

$$z_j = \frac{Q_j(P(z))}{R_j(P(z))}.$$

Equivalently,

$$\boxed{Q_j(P(z)) = z_j R_j(P(z))}.$$

We now show that R_j does not vanish on $P(\mathbb{C}^n)$. Suppose that $w = P(z) \in P(\mathbb{C}^n)$ and $R_j(w) = 0$. Then the preceding polynomial identity gives $Q_j(w) = z_j R_j(w) = 0$. Hence, on the open set $P(\mathbb{C}^n) \subseteq \mathbb{C}^n$, the zero set of R_j is contained in the zero set of Q_j . But $P(\mathbb{C}^n)$ is open by the preceding corollary, and Q_j and R_j are relatively prime. By elementary algebraic geometry, such a containment cannot occur on a non-empty open set unless R_j has no zero there. Therefore

$$R_j(w) \neq 0 \quad \text{for every } w \in P(\mathbb{C}^n).$$

It follows that $R_j(P(z))$ is a polynomial on $\mathbb{C}^n$ with no zeros. Hence, by the Nullstellensatz, it must be a non-zero constant. After rescaling Q_j and R_j , we may assume that $R_j(P(z)) = 1$. Thus $z_j = Q_j(P(z))$ for every $j = 1, \dots, n$. Putting $Q = (Q_1, \dots, Q_n)$, we obtain

$$\boxed{Q(P(z)) = z.}$$

Now let $w \in P(\mathbb{C}^n)$. Then $w = P(z)$ for some z , and therefore $P(Q(w)) = P(Q(P(z))) = P(z) = w$. Hence $P(Q(w)) = w$ for every w in the non-empty open set $P(\mathbb{C}^n)$.

But both

$$w \mapsto P(Q(w)) \quad \text{and} \quad w \mapsto w$$

are polynomial maps. Since they agree on a non-empty open subset of $\mathbb{C}^n$, they agree everywhere. Thus

$$\boxed{P(Q(w)) = w}$$

for every $w \in \mathbb{C}^n$.

Therefore P is bijective, with polynomial inverse Q . □

Remark 2. Rudin's proof actually gives the stronger statement that if a polynomial map

$$P : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

is injective, then it is a polynomial automorphism. That is, P is bijective and its inverse

$$P^{-1} : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

is itself a polynomial map.

Proof via Finite Fields

Let F be a finite field.

Lemma 2. *Let F be a finite field and let*

$$P : F^n \longrightarrow F^n$$

be a polynomial map. If P is injective, then P is bijective.

Proof. Any injection from a finite set to itself is necessarily bijective. Notice that no polynomial assumption is needed here! □

Notation. As an abuse of notation, we may sometimes write

$$P(x_1, \dots, x_n) \in F[x_1, \dots, x_n]$$

as $P(\mathbf{x})$, and

$$P(x_1, \dots, x_n, y_1, \dots, y_n) \in F[x_1, \dots, x_n, y_1, \dots, y_n]$$

as $P(\mathbf{x}, \mathbf{y})$.

Now we want to extend the above observation to

$$\overline{F}^n.$$

For this, we shall need a couple of versions of Hilbert's Nullstellensatz.

Version I (Hilbert's Nullstellensatz). Let V be an algebraic variety over an algebraically closed field $\mathbb{K}$. Then

$$I(V(I)) = \sqrt{I},$$

where I is the ideal associated with the variety, and

$$\sqrt{I} = \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f^r \in I \text{ for some } r \in \mathbb{N}\}.$$

Since $\mathbb{K}[x_1, \dots, x_n]$ is Noetherian, we may write

$$I = (g_1, \dots, g_r).$$

Then

$$V(I) = \{a \in \mathbb{K}^n \mid g_i(a) = 0 \text{ for every } i = 1, \dots, r\},$$

and

$$I(V(I)) = \{h \in \mathbb{K}[x_1, \dots, x_n] \mid h(a) = 0 \text{ for every } a \in V(I)\}.$$

Thus h vanishes at exactly all the points at which $g_1, \dots, g_r$ vanish. By the Nullstellensatz,

$$I(V(I)) = \sqrt{I}.$$

Hence, for some $N \geq 1$, $h^N \in I$. Therefore there exist $Q_1, \dots, Q_r \in \mathbb{K}[x_1, \dots, x_n]$ such that

$$h^N = \sum_{i=1}^r g_i Q_i.$$

This gives us the second version of the Nullstellensatz which we shall use.

Version II (Hilbert's Nullstellensatz). Let $\mathbb{K}$ be an algebraically closed field. Suppose $g_1, \dots, g_r, h \in \mathbb{K}[x_1, \dots, x_n]$ and suppose that h vanishes at every point at which $g_1, \dots, g_r$ vanish. Then there exist $Q_1, \dots, Q_r \in \mathbb{K}[x_1, \dots, x_n]$ and $N \geq 1$ such that

$$h^N = \sum_{i=1}^r g_i Q_i.$$

We may interpret it as saying that if an algebraic problem is insoluble, then there exists a *finitary algebraic obstruction* witnessing this lack of a solution. Suppose an algebraic problem is insoluble that is there exists atleast one $b \in \mathbb{K}^n$ such that $P = b$ has no solution. Thus there doesn't exist a common sloution for $\{P_i - b_i\}_{i=1}^n$, where $b = (b_1, \dots, b_n)$. Now by Nullstellensatz we get the ideal generated by these polynomials is the whole ring. Thus we get a finite number of polynomials witnessing this lack of solutions. The main motivation for the Nullstellensatz here is that it allows us to translate properties such as injectivity and surjectivity of polynomial maps into algebraic statements involving only a finite amount of data. We will understand this with the help of next two lemmas. We will see why we say this in a while, in the course of our proof.

Lemma 3. *Let*

$$P : F^n \longrightarrow F^n, \quad P = (P_1, \dots, P_n),$$

be a polynomial map, where F is algebraically closed. Then P is injective if and only if there exist polynomials $Q_{i,j} \in F[x_1, \dots, x_n, y_1, \dots, y_n]$, $1 \leq i, j \leq n$, and an integer $N_j \geq 1$ such that

$$(x_j - y_j)^{N_j} = \sum_{i=1}^n (P_i(\mathbf{x}) - P_i(\mathbf{y})) Q_{i,j}(\mathbf{x}, \mathbf{y})$$

for every $j = 1, \dots, n$.

Proof. Suppose first that such identities hold. If

$$P(a_1, \dots, a_n) = P(b_1, \dots, b_n),$$

then

$$P_i(\mathbf{a}) - P_i(\mathbf{b}) = 0 \quad \text{for every } i.$$

Substituting $(\mathbf{a}, \mathbf{b})$ into the above identity gives

$$(a_j - b_j)^{N_j} = 0 \quad \text{for every } j.$$

Since F is a field, $a_j = b_j, \forall j$. Hence $\mathbf{a} = \mathbf{b}$, and therefore P is injective.

Conversely, suppose that P is injective. We need to prove the above identities.

Fix $j \in \{1, \dots, n\}$. Observe that the polynomial $x_j - y_j$ vanishes at every common zero of the collection $\{P_i(\mathbf{x}) - P_i(\mathbf{y})\}_{i=1}^n$. Indeed, if $P_i(\mathbf{x}) - P_i(\mathbf{y}) = 0, \forall i$. Then $P(\mathbf{x}) = P(\mathbf{y})$, and injectivity gives $\mathbf{x} = \mathbf{y}$. In particular, $x_j - y_j = 0$.

Hence, by Hilbert's Nullstellensatz, there exists $N_j \geq 1$ such that

$$(x_j - y_j)^{N_j} \in (P_1(\mathbf{x}) - P_1(\mathbf{y}), \dots, P_n(\mathbf{x}) - P_n(\mathbf{y})).$$

Thus there exist polynomials $Q_{i,j}$ such that

$$(x_j - y_j)^{N_j} = \sum_{i=1}^n (P_i(\mathbf{x}) - P_i(\mathbf{y})) Q_{i,j}(\mathbf{x}, \mathbf{y}).$$

This proves the result. □

Similarly, we can express the failure of surjectivity algebraically.

Lemma 4. *Let*

$$P : F^n \longrightarrow F^n$$

be a polynomial map, where F is algebraically closed. Then P is not surjective if and only if there exist

$$\mathbf{z}_0 = (z_1, \dots, z_n) \in F^n$$

and polynomials

$$R_1, \dots, R_n \in F[x_1, \dots, x_n]$$

such that

$$\sum_{i=1}^n R_i(\mathbf{x}) (P_i(\mathbf{x}) - z_i) = 1.$$

Proof. Suppose first that the displayed identity holds. Then there cannot exist $\mathbf{a} \in F^n$ such that $P(\mathbf{a}) = \mathbf{z}_0$. Indeed, substituting such an $\mathbf{a}$ into the identity would give $1 = 0$. Thus $\mathbf{z}_0 \notin P(F^n)$, so P is not surjective.

Conversely, suppose that P is not surjective. Then there exists

$$\mathbf{z}_0 = (z_1, \dots, z_n) \in F^n$$

such that

$$P(\mathbf{x}) \neq \mathbf{z}_0 \quad \text{for every } \mathbf{x} \in F^n.$$

Consequently, the polynomials

$$P_1(\mathbf{x}) - z_1, \dots, P_n(\mathbf{x}) - z_n$$

have no common zero in F^n . By Hilbert's Nullstellensatz,

$$(P_1(\mathbf{x}) - z_1, \dots, P_n(\mathbf{x}) - z_n) = F[x_1, \dots, x_n].$$

Hence 1 belongs to this ideal, so there exist polynomials $R_1, \dots, R_n$ such that

$$\sum_{i=1}^n R_i(\mathbf{x})(P_i(\mathbf{x}) - z_i) = 1.$$

This proves the result. □

These two lemmas justify our earlier interpretation of the Nullstellensatz: it allows us to translate geometric properties such as injectivity and non-surjectivity of polynomial maps into *finite algebraic data*.

Theorem 2. (*Ax–Grothendieck*) *Let $\mathbb{K}$ be an algebraically closed field, and let*

$$P : \mathbb{K}^n \longrightarrow \mathbb{K}^n$$

be an injective polynomial map. Then P is surjective.

Proof. We first consider the case where $\mathbb{K}$ is the algebraic closure of a finite field k . From [Lemma 3] and [Lemma 4], we obtain a finite collection of polynomials $\{P_i, Q_{i,j}, R_i\}$ together with the point $\mathbf{z}_0 = (z_1, \dots, z_n)$, which witness, respectively, the injectivity and the failure of surjectivity of P .

Let C be the finite collection of all coefficients occurring in these polynomials and in $\mathbf{z}_0$. Since C is a finite subset of $\bar{k}$, consider the field $k(C)$ generated by C over k . These coefficients are contained in some finite extension of $k \subseteq \mathbb{K}$ we get that the extension $k(C)$ over k is finite. Since k is finite, it follows that $k(C)$ is also a finite field.

All the coefficients occurring in $P_i, Q_{i,j}, R_i$, and $\mathbf{z}_0$ lie in $k(C)$. Hence our polynomial map restricts to

$$P : k(C)^n \longrightarrow k(C)^n.$$

Moreover, the polynomial identities from [Lemma 3] and [Lemma 4] still hold over $k(C)$. Consequently, this map is injective but not surjective. But $k(C)^n$ is a finite set, and every injective self-map of a finite set is surjective. This is a contradiction.

Hence the theorem holds over the algebraic closure of a finite field.

We now consider an arbitrary algebraically closed field $\mathbb{K}$. Again, collect all the coefficients occurring in

$$P_i, \quad Q_{i,j}, \quad R_i, \quad \text{and} \quad \mathbf{z}_0.$$

Let $C = \{\alpha_1, \dots, \alpha_r\} \subseteq K$ be the finite collection of all coefficients occurring in our polynomial identities. Let A be the subring of $\mathbb{K}$ generated by C over the prime subring of $\mathbb{K}$. Thus

$$A = \begin{cases} \mathbb{Z}[\alpha_1, \dots, \alpha_r], & \text{if } \text{char}(\mathbb{K}) = 0, \\ \mathbb{F}_p[\alpha_1, \dots, \alpha_r], & \text{if } \text{char}(\mathbb{K}) = p > 0. \end{cases}$$

Choose a maximal ideal $\mathfrak{m} \subset A$ and put $\kappa = A/\mathfrak{m}$. Since $\mathfrak{m}$ is maximal, κ is a field. Reducing all coefficients modulo $\mathfrak{m}$, we obtain a polynomial map

$$\overline{P} : \kappa^n \longrightarrow \kappa^n.$$

It's easy to check that the identities from [Lemma 3] and [Lemma 4] survive reduction modulo $\mathfrak{m}$. Hence $\overline{P}$ is injective but not surjective. Thus it remains only to prove that κ is finite. When $A = \mathbb{F}_p[\alpha_1, \dots, \alpha_r]$ we have already seen that κ is a finite field, only thing left to show is when $A = \mathbb{Z}[\alpha_1, \dots, \alpha_r]$.

Claim 0.1. *The field*

$$\kappa = A/\mathfrak{m}$$

is finite.

Proof. First we show that

$$\text{char}(\kappa) \neq 0.$$

Suppose, for contradiction, that

$$\text{char}(\kappa) = 0.$$

Writing $\overline{\alpha}_i$ for the image of α_i in κ , we have

$$\kappa = \mathbb{Z}[\overline{\alpha}_1, \dots, \overline{\alpha}_r].$$

Since κ has characteristic 0, it contains $\mathbb{Q}$. Therefore

$$\kappa = \mathbb{Q}(\overline{\alpha}_1, \dots, \overline{\alpha}_r) = \mathbb{Q}[\overline{\alpha}_1, \dots, \overline{\alpha}_r].$$

By Zariski's Lemma [Atiyah], $[\kappa : \mathbb{Q}] < \infty$. In particular, each $\overline{\alpha}_i$ is algebraic over $\mathbb{Q}$. For each i , choose a monic polynomial

$$X^{d_i} + a_{i,d_i-1}X^{d_i-1} + \dots + a_{i,0} \in \mathbb{Q}[X]$$

satisfied by $\overline{\alpha}_i$. Only finitely many rational numbers occur among the coefficients $a_{i,j}$. Hence we may choose an integer $N \neq 0$ containing all their denominators and put

$$B = \mathbb{Z}\left[\frac{1}{N}\right].$$

Then $a_{i,j} \in B$ for every i, j , so each $\overline{\alpha}_i$ is integral over B . Consequently, $B[\overline{\alpha}_1, \dots, \overline{\alpha}_r]$ is integral over B . But since

$$\kappa = \mathbb{Z}[\overline{\alpha}_1, \dots, \overline{\alpha}_r]$$

and $\frac{1}{N} \in \kappa$, we have

$$B[\overline{\alpha}_1, \dots, \overline{\alpha}_r] = \kappa.$$

Thus κ is integral over B . Now κ is a field. We claim that this forces B to be a field.

Indeed, take $0 \neq b \in B$. Since κ is a field, $b^{-1} \in \kappa$. Since κ/B is integral, b^{-1} is integral over B . Hence

$$(b^{-1})^m + c_{m-1}(b^{-1})^{m-1} + \dots + c_0 = 0, \quad c_i \in B.$$

Multiplying by b^{m-1} , we obtain

$$b^{-1} = -(c_{m-1} + c_{m-2}b + \dots + c_0b^{m-1}) \in B.$$

Thus every nonzero element of B is invertible, so B is a field. But

$$B = \mathbb{Z}\left[\frac{1}{N}\right]$$

is not a field. Indeed, choose a prime $q \nmid N$. Then

$$\frac{1}{q} \notin B.$$

This is a contradiction.

Remark 3. There is another way to finish the contradiction above, using Noetherian modules. Recall that, under the assumption

$$\text{char}(\kappa) = 0,$$

we obtained that each $\bar{\alpha}_i$ is integral over B .

Therefore

$$\kappa = B[\bar{\alpha}_1, \dots, \bar{\alpha}_r]$$

is a finitely generated B -module. Indeed, if an element α satisfies a monic polynomial

$$\alpha^d + b_{d-1}\alpha^{d-1} + \dots + b_0 = 0, \quad b_i \in B,$$

then every higher power of α can be expressed as a B -linear combination of

$$1, \alpha, \dots, \alpha^{d-1}.$$

Thus $B[\alpha]$ is finitely generated as a B -module. Applying this successively to the finitely many integral elements $\bar{\alpha}_1, \dots, \bar{\alpha}_r$, we conclude that

$$\kappa = B[\bar{\alpha}_1, \dots, \bar{\alpha}_r]$$

is finitely generated as a B -module. Now $B = \mathbb{Z}[1/N]$ is Noetherian, since it is a localization of the Noetherian ring $\mathbb{Z}$ at $\{1, N, N^2, \dots\}$. Hence the finitely generated B -module κ is a Noetherian B -module.

But, since $\text{char}(\kappa) = 0$ and κ is a field, we have $\mathbb{Q} \subseteq \kappa$. Moreover, $\mathbb{Q}$ is a B -submodule of κ . Since every submodule of a Noetherian module is finitely generated, it follows that $\mathbb{Q}$ would be a Noetherian B -module. We now show that this is impossible. Choose distinct primes

$$q_1, q_2, q_3, \dots$$

none of which divides N . Consider the B -submodules of $\mathbb{Q}$

$$B \subset B\left[\frac{1}{q_1}\right] \subset B\left[\frac{1}{q_1}, \frac{1}{q_2}\right] \subset B\left[\frac{1}{q_1}, \frac{1}{q_2}, \frac{1}{q_3}\right] \subset \dots$$

Every inclusion is strict. Indeed,

$$\frac{1}{q_{r+1}} \notin B\left[\frac{1}{q_1}, \dots, \frac{1}{q_r}\right],$$

since the denominators of elements of the latter ring can involve only primes dividing

$$Nq_1 \cdots q_r,$$

whereas q_{r+1} divides none of these. Thus we have an infinite strictly ascending chain of B -submodules of $\mathbb{Q}$:

$$B \subsetneq B\left[\frac{1}{q_1}\right] \subsetneq B\left[\frac{1}{q_1}, \frac{1}{q_2}\right] \subsetneq \dots$$

Hence $\mathbb{Q}$ is not a Noetherian B -module, a contradiction.

Therefore

$$\boxed{\text{char}(\kappa) \neq 0.}$$

Since κ is a field, its characteristic must therefore be a prime p . Equivalently, $\mathfrak{m} \cap \mathbb{Z} = (p)$. Hence $\kappa = \mathbb{F}_p[\bar{\alpha}_1, \dots, \bar{\alpha}_r]$.

Now κ is a field finitely generated as an $\mathbb{F}_p$ -algebra. By Zariski's Lemma [Atiyah], $[\kappa : \mathbb{F}_p] < \infty$. Therefore $|\kappa| = p^{[\kappa : \mathbb{F}_p]} < \infty$.

Thus κ is a finite field.

□

Returning to

$$\overline{P} : \kappa^n \longrightarrow \kappa^n,$$

we have obtained an injective but non-surjective self-map of the finite set κ^n , which is impossible. Therefore P must be surjective. □

Remark 4. At first one might try to argue directly that

$$\kappa = \mathbb{Z}[\overline{\alpha}_1, \dots, \overline{\alpha}_r]$$

is a Noetherian $\mathbb{Z}$ -module. This is not justified: being finitely generated as a $\mathbb{Z}$ -algebra does not imply being finitely generated as a $\mathbb{Z}$ -module. For example, $\mathbb{Z}[x]$ is generated by one element as a $\mathbb{Z}$ -algebra, but as a $\mathbb{Z}$ -module it requires the infinitely many generators

$$1, x, x^2, x^3, \dots$$

The role of Zariski's Lemma and the passage to

$$B = \mathbb{Z} \left[\frac{1}{N} \right]$$

is precisely to repair this problem. Zariski's Lemma first tells us that the $\overline{\alpha}_i$'s are algebraic over $\mathbb{Q}$; after clearing finitely many denominators they become integral over B . Integrality of finitely many generators then upgrades “finite as an algebra” to

$$\boxed{\text{finite as a } B\text{-module.}}$$

Only at this point can the Noetherian-module argument be applied.

The general case when X is an algebraic variety over an algebraically closed field $\mathbb{K}$ is similar (see [EGA]). When $\mathbb{K} = \mathbb{C}$ a topological proof of the general theorem was given by Borel in 1969 (see [Borel]). The technique we used to replace variety defined over $\mathbb{K}$ to some finitely generated $\mathbb{Z}$ algebra A is called spreading out, its properties are discussed in details in [EGA]

Theorem 3. *Let G be a finite p -group acting algebraically on $\mathbb{A}_k^n$, where k is an algebraically closed field with*

$$\text{char}(k) \neq p.$$

Then the action of G has a fixed point.

Proof. A finite group acting algebraically on a variety means that G acts on the underlying set and also the action is a morphism of varieties. This is equivalent to giving a homomorphism from G to $\text{Aut}_k(X)$. We will follow [Serre] for the proof.

We first prove the theorem when $k = \overline{\mathbb{F}_\ell}$, where $\ell \neq p$ is a prime.

Since G is finite and the action is algebraic, for every $g \in G$ we may write

$$g(x_1, \dots, x_n) = (P_{g,1}(x_1, \dots, x_n), \dots, P_{g,n}(x_1, \dots, x_n)),$$

where

$$P_{g,i} \in \overline{\mathbb{F}_\ell}[x_1, \dots, x_n].$$

Only finitely many coefficients occur among all the polynomials $P_{g,i}$, since G is finite. Hence all these coefficients are contained in some finite extension $\mathbb{F}_{\ell^m} \subseteq \overline{\mathbb{F}_\ell}$. Thus the action of G is defined over $\mathbb{F}_{\ell^m}$, and we obtain an action of G on the finite set $\mathbb{A}^n(\mathbb{F}_{\ell^m}) = \mathbb{F}_{\ell^m}^n$. Now $|\mathbb{F}_{\ell^m}^n| = \ell^{mn}$, which is not divisible by p , since $\ell \neq p$.

On the other hand, every orbit of a p -group has cardinality a power of p . Indeed, for $x \in \mathbb{F}_{\ell^m}^n$,

$$|G \cdot x| = [G : G_x],$$

and this divides $|G|$, which is a power of p . If there were no fixed point, every orbit would therefore have cardinality divisible by p . Since $\mathbb{F}_{\ell^m}^n$ is the disjoint union of its orbits, this would imply

$$p \mid |\mathbb{F}_{\ell^m}^n| = \ell^{mn},$$

which is impossible. Hence G has a fixed point in $\mathbb{F}_{\ell^m}^n$, and therefore in $\mathbb{A}_{\mathbb{F}_\ell}^n$.

We now reduce the general case to the finite-field case. Suppose, towards a contradiction, that the action of G on $\mathbb{A}_k^n$ has no fixed point. Since G is finite, only finitely many coefficients occur in the polynomials $P_{g,i}$ defining the action. Let $\Lambda \subseteq k$ be the subring generated by these coefficients over the prime ring of k .

If $\text{char}(k) = 0$, we may regard Λ as a finitely generated $\mathbb{Z}$ -algebra. If $\text{char}(k) = \ell > 0$, it is a finitely generated $\mathbb{F}_\ell$ -algebra. A point

$$x = (x_1, \dots, x_n) \in k^n$$

is fixed by G precisely when

$$x_i - P_{g,i}(x_1, \dots, x_n) = 0$$

for every $g \in G$ and every $1 \leq i \leq n$.

By assumption, this finite system of polynomial equations has no solution in k^n . Since k is algebraically closed, Hilbert's Nullstellensatz gives polynomials

$$Q_{g,i} \in k[x_1, \dots, x_n]$$

such that

$$\sum_{g \in G} \sum_{i=1}^n (x_i - P_{g,i}(x_1, \dots, x_n)) Q_{g,i}(x_1, \dots, x_n) = 1. \quad (1)$$

We now enlarge Λ , if necessary, so that it contains all the coefficients of the polynomials $Q_{g,i}$. We also arrange that p is invertible in Λ . In characteristic 0, this is done by adjoining $1/p$:

$$\Lambda \longmapsto \Lambda \left[\frac{1}{p} \right].$$

In positive characteristic $\ell \neq p$, p is already nonzero and hence invertible in the prime field $\mathbb{F}_\ell$.

Thus Λ is still a finitely generated ring, all the polynomials occurring in the action and in (1) are defined over Λ , and

$$\frac{1}{p} \in \Lambda$$

in the characteristic-zero case.

Choose a maximal ideal $\mathfrak{m} \subseteq \Lambda$ and put $F = \Lambda/\mathfrak{m}$.

The field F is finite. Indeed, in positive characteristic this follows immediately from Zariski's Lemma [Atiyah], since F is a field finitely generated over a finite field. In characteristic 0, this follows from our [Theorem 2] of our [Theorem 2].

Moreover, $\text{char}(F) \neq p$. Indeed, p is invertible in Λ , and therefore its image remains invertible in F . In particular, its image cannot be zero. We now reduce all the polynomials defining the action modulo $\mathfrak{m}$. Denote the resulting polynomials by $\overline{P}_{g,i} \in F[x_1, \dots, x_n]$. The polynomial identities expressing the group law $P_g \circ P_h = P_{gh}$ also survive reduction. Hence these reduced polynomials define an algebraic action $G \curvearrowright \mathbb{A}_F^n$.

Similarly, reducing (1) modulo $\mathfrak{m}$, we obtain

$$\sum_{g \in G} \sum_{i=1}^n (x_i - \overline{P}_{g,i}(x)) \overline{Q}_{g,i}(x) = 1. \quad (2)$$

This identity remains valid after extending scalars from F to its algebraic closure $\overline{F}$.

If the reduced action had a fixed point $a = (a_1, \dots, a_n) \in \overline{F}^n$, then $a_i - \overline{P}_{g,i}(a) = 0$ for every $g \in G$ and every i . Substituting a into (2) would give $0 = 1$, a contradiction. Thus the induced action $G \curvearrowright \mathbb{A}_{\overline{F}}^n$ has no fixed point.

But F is a finite field of characteristic different from p . Therefore $\overline{F}$ is the algebraic closure of a finite field of characteristic different from p , and the first part of the proof shows that every algebraic action of G on $\mathbb{A}_{\overline{F}}^n$ has a fixed point. This is a contradiction. Hence the original action of G on $\mathbb{A}_k^n$ must have a fixed point. $\square$

As a corollary of the previous theorem we get the next theorem.

Theorem 4. *Let σ be an automorphism of the complex affine n -space $\mathbb{C}^n$, viewed as an algebraic variety. Assume that $\sigma^2 = 1$. Then σ has a fixed point.*

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